



**Revised Generation Interconnection
System Impact Study Report
for
Queue Project AE2-047
RIDGEWAY-SOLITE 69 KV
30 MW Capacity / 50 MW Energy**

February, 2020

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1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances, a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection, may also contribute to the need for the same network reinforcement. Cost allocation rules for network upgrades can be found in PJM Manual 14A, Attachment B. The possibility of sharing the reinforcement costs with other projects may be identified in the feasibility study, but the actual allocation will be deferred until the impact study is performed.

The Interconnection Customer seeking to interconnect a wind generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

PJM utilizes manufacturer models to ensure the performance of turbines is properly captured during the simulations performed for stability verification and, where applicable, for compliance with low voltage ride through requirements. Turbine manufacturers provide such models to their customers. The list of manufacturer models PJM has already validated is contained in Attachment B of Manual 14G. Manufacturer models may be updated from time to time, for various reasons such as to reflect changes to the control systems or to more accurately represent the capabilities turbines and controls which are currently available in the field. Additionally, as new turbine models are developed, turbine manufacturers provide such new models which must be used in the conduct of these studies. PJM needs adequate time to evaluate the new models in order to reduce delays to the System Impact Study process timeline for the Interconnection Customer as well as other Interconnection Customers in the study group. Therefore, PJM will require that any Interconnection Customer with a new manufacturer model must supply that model to PJM, along with a \$10,000 fully refundable deposit, no later than three (3) months prior to the starting date of the System Impact Study (See Section 4.3 for starting dates) for the Interconnection Request which shall specify the use of the new model. The Interconnection Customer will be required to submit a completed dynamic model study request form (Attachment B-1 of Manual 14G) in order to document the request for the study.

2 General

The Interconnection Customer has proposed to install a Solar generating facility located in Henry County, Virginia (See Figure 2). The installed facilities will have a total capability of 50 MW with 30 MW of this output being recognized by PJM as Capacity (the approved Capacity is 32.4 MW, but it was modeled as 30 MW in this study). The Point of Interconnection will be to the Ridgeway – Solite 69 kV section of the Ridgeway – Corning Glass 69 kV circuit (See Figure 1).

The proposed in-service date for this project is December 15, 2021. This study does not imply AEP's commitment to this in-service date.

The objective of this System Impact Study is to determine budgetary cost estimates and approximate construction timelines for identified transmission facilities required to connect the proposed generating facilities to the AEP transmission system. These reinforcements include the Attachment Facilities, Local Upgrades, and Network Upgrades required for maintaining the reliability of the AEP transmission system.

Queue Number	AE2-047
Project Name	RIDGEWAY-SOLITE 69 KV
State	Virginia
County	Henry
Transmission Owner	AEP
MFO	50
MWE	50
MWC	30
Fuel	Solar
Base case Study Year	2022

2.1 Point of Interconnection

AE2-047 will interconnect with the AEP transmission system at the Ridgeway – Solite 138 kV section of the Ridgeway – Corning Glass 138 kV circuit.

To accommodate the interconnection on the Ridgeway - Solite 69 kV line, a new three (3) circuit breaker 69 kV switching station physically configured in a breaker and half bus arrangement but operated as a ring-bus will be constructed (see Figure 1). Installation of associated protection and control equipment, 69 kV line risers, SCADA, and 69 kV revenue metering will also be required. AEP reserves the right to specify the final acceptable configuration considering design practices, future expansion, and compliance requirements.

2.2 Cost Summary

The AE2-047 project will be responsible for the following costs:

Based upon the results of this Impact Study, the construction of the 50 MW (30 MW Capacity) solar generating facility of Michaux Solar Center (PJM Project #AE2-047) will require the following additional interconnection charges. This plan of service will interconnect the proposed generating facility in a manner that will provide operational reliability and flexibility to both the AEP system and the Michaux Solar Center generating facility.

Description	Total Cost
Attachment Facilities	\$500,000
Direct Connection Network Upgrade	\$4,300,000
Non Direct Connection Network Upgrades	\$1,100,000
Allocation for New System Upgrades	\$0
Contribution for Previously Identified Upgrades	\$0
Total Costs	\$5,900,000

The estimates provided in this report are preliminary in nature, as they were determined without the benefit of detailed engineering studies. Final estimates will require an on-site review and coordination to determine final construction requirements.

3 Transmission Owner Scope of Work

4 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
69 kV Revenue Metering	\$200,000
Generator lead first span exiting the POI station, including the first structure outside the fence	\$300,000
Total Attachment Facility Costs	\$500,000

5 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Construct a new three (3) circuit breaker 69 kV switching station physically configured in a breaker and half bus arrangement but operated as a ring-bus. Installation of associated protection and control equipment, 69 kV line risers and SCADA will also be required.	\$4,350,000
Total Direct Connection Facility Costs	\$4,350,000

6 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Ridgeway – Solite 69 kV T-Line Cut in	\$700,000
Upgrade line protection and controls at the Ridgeway 69 kV substation	\$200,000
Upgrade line protection and controls at the Corning Glass 69 kV substation.	\$200,000
Total Non-Direct Connection Facility Costs	\$1,100,000

7 Incremental Capacity Transfer Rights (ICTRs)

Will be determined at a later study phase

8 Schedule

It is anticipated that the time between receipt of executed Agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would generally be between 24 to 36 months after signing Agreement execution.

9 Interconnection Customer Requirements

It is understood that the Interconnection Customer is responsible for all costs associated with this interconnection. The costs above are reimbursable to AEP. The cost of the Interconnection Customer's generating plant and the costs for the line connecting the generating plant to the transmission system are not included in this report; these are assumed to be the Interconnection Customer's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System" document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

11 Network Impacts

The Queue Project AE2-047 was evaluated as a 50.0 MW (Capacity 32.4 MW) injection into a tap of the Ridgeway – Solite 69 kV line in the AEP area. Project AE2-047 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-047 was studied with a commercial probability of 1.00. Potential network impacts were as follows:

Summer Peak Load Flow

12 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

13 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

14 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

15 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
6845538	242555	05BLAINE	138.0	AEP	242773	05ROANO1	138.0	AEP	1	24749905SMITHMTN2 - 24280205SMITHMTN1 CKT Z1	operation	223.0	101.28	103.27	AC	4.87

16 Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

To be determined in the Facilities Study Phase

17 Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase

18 Light Load Analysis

Light Load Studies (applicable to wind, coal, nuclear, and pumped storage projects).

None.

19 System Reinforcements

None

Affected Systems

20 Affected Systems

20.1 TVA

None

20.2 Duke Energy Progress

Duke Energy Progress Impacts to be determined during later study phases (as applicable).

20.3 MISO

MISO Impacts to be determined during later study phases (as applicable).

20.4 LG&E

None

21 Contingency Descriptions

Contingency Name	Contingency Definition
AEP_P7-1_#10781-B	CONTINGENCY 'AEP_P7-1_#10781-B' OPEN BRANCH FROM BUS 936160 TO BUS 304024 CKT 1 / 936160 AD2-022 TAP 230 304024 6ROXSEP230 T 230 1 OPEN BRANCH FROM BUS 936170 TO BUS 304094 CKT 1 / 936170 AD2-023 TAP 230 304094 6YANCY TAP 230 1 OPEN BRANCH FROM BUS 304098 TO BUS 304094 CKT 1 / 304098 6CONCRD230TT 230 304094 6YANCY TAP 230 1 OPEN BRANCH FROM BUS 304094 TO BUS 304095 CKT 1 / 304094 6YANCY TAP 230 304095 6YANCYVILLE 230 1 END
AEP_P7-1_#10781-A	CONTINGENCY 'AEP_P7-1_#10781-A' OPEN BRANCH FROM BUS 242530 TO BUS 936160 CKT 1 / 242530 05EDANV1 230 936160 AD2-022 TAP 230 1 OPEN BRANCH FROM BUS 242531 TO BUS 936170 CKT 1 / 242531 05EDANV2 230 936170 AD2-023 TAP 230 1 END
AEP_P4_#10171_05AXTON 138_G	CONTINGENCY 'AEP_P4_#10171_05AXTON 138_G' OPEN BRANCH FROM BUS 242509 TO BUS 242514 CKT 1 / 242509 05AXTON 765 242514 05J.FERR 765 1 OPEN BRANCH FROM BUS 242509 TO BUS 242545 CKT 1 / 242509 05AXTON 765 242545 05AXTONX 138 1 OPEN BRANCH FROM BUS 242544 TO BUS 242545 CKT SR / 242544 05AXTON 138 242545 05AXTONX 138 SR OPEN BRANCH FROM BUS 242544 TO BUS 242545 CKT ZB / 242544 05AXTON 138 242545 05AXTONX 138 ZB OPEN BRANCH FROM BUS 242544 TO BUS 242619 CKT 2 / 242544 05AXTON 138 242619 05DANVL1 138 2 END
247499 05SMITHMTN2 - 242802 05SMITHMTN1 CKT Z1	CONTINGENCY '247499 05SMITHMTN2 - 242802 05SMITHMTN1 CKT Z1' /MISSING SINGLE DISCONNECT BRANCH FROM BUS 247499 TO BUS 242802 CKT Z1 END

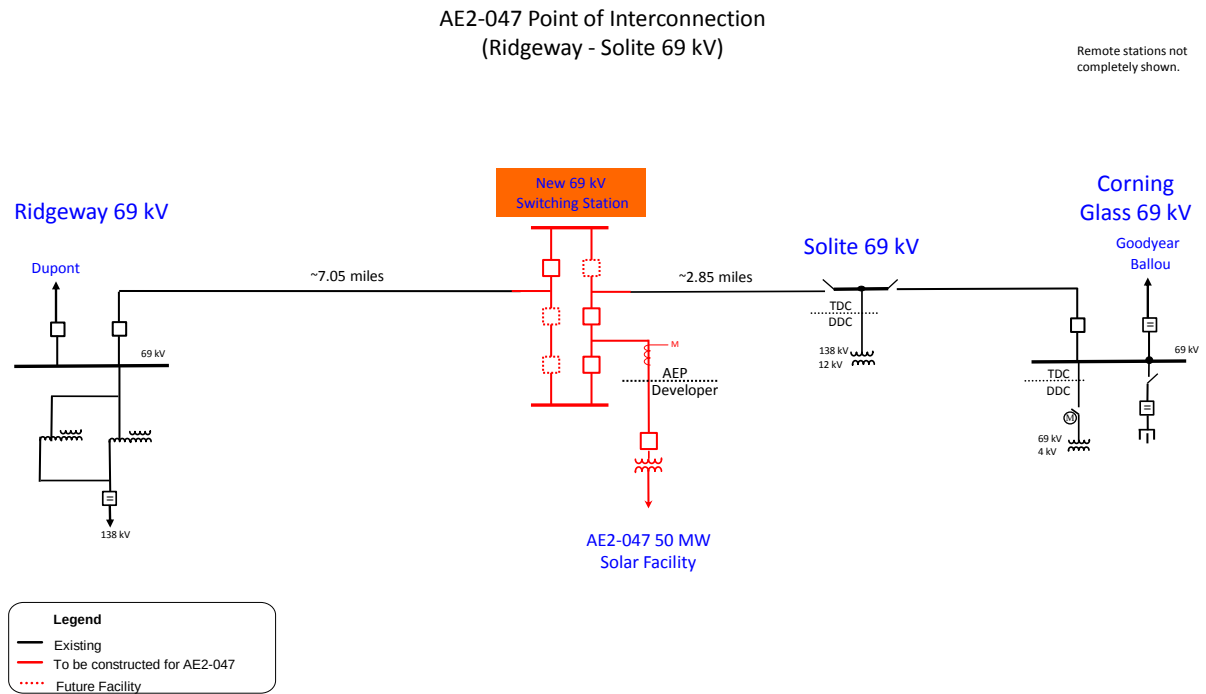
Short Circuit

22 Short Circuit

The following Breakers are overduty

Bus Number	Bus Name	BREAKER	Type	Capacity (Amps)	Duty Percentage Post Queue	Duty Percentage Pre Queue

23 Figure 1: AE2-047 Point of Interconnection (Ridgeway - Solite 69 kV) Single-Line Diagram



24 Figure 1: AE2-047 Point of Interconnection (Ridgeway - Solite 69 kV)

